

### Examples of Medical Prefixes, Suffixes and Word roots

| Prefixes  | Meanings                     | Examples - Definitions                                                                                                                 |
|-----------|------------------------------|----------------------------------------------------------------------------------------------------------------------------------------|
| a-, an-   | without, lack of             | anuria - lack of urine output                                                                                                          |
| ab-       | away from                    | abnormal - a structure or process that is not normal<br>abduction - movement of a limb or other part away from the midline of the body |
| ad-       | toward, near                 | adrenal glands - two small triangular endocrine glands situated upon the upper end of each kidney                                      |
| ambi-     | both sides                   | ambidextrous - using both hands                                                                                                        |
| ante-     | before, forward              | antenatal – pre-birth                                                                                                                  |
| anti-     | against                      | Antidotes - a therapeutic substance used to counteract the toxic actions of a specific substance                                       |
| apo-      | off, away from               | apophysis - growth or protuberance                                                                                                     |
| auto-     | self                         |                                                                                                                                        |
| bi-       | two, both                    | bilateral - occurring on both sides of the body                                                                                        |
| cata-     | downwards                    | Catabolism - the process of breaking down complex chemicals into simple chemicals                                                      |
| con-      | with, together               | congenital - disease or physical abnormality present from birth                                                                        |
| de-       | without                      | depigmentation - without pigment                                                                                                       |
| diplo-    | double                       | diplopia - double vision                                                                                                               |
| dys-      | painful, difficult, abnormal | Dyspnoea - difficulty in breathing                                                                                                     |
| ec-, ecto | out, outside                 | ectoderm - the outer layer of an early embryo                                                                                          |
| endo-     | within, inside               | endoscopy - an examination of the inside of the body using an endoscope                                                                |
| epi-      | above                        | epigastric - above the stomach                                                                                                         |
| eu-       | normal                       | euthyroid - normal thyroid function                                                                                                    |
| ex-       | outwards                     | exostosis - condition of outward, or projecting, bone                                                                                  |
| extra-    | outside of                   | extrapleural - outside the pleural cavity                                                                                              |
| hemi-     | half                         | hemiplegia - paralysis that is limited to one side of the body                                                                         |

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|-----------|----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| hetero-   | different                  | heterograft - A transplant from one animal to another of a different species                                                                                               |
| homo-     | same                       | homoplasty - surgery to replace lost tissues by grafting similar tissues from another person                                                                               |
| hyper-    | excessive, too much, above | hyperplasia - an abnormal increase in the number of cells in a tissue.                                                                                                     |
| hypo-     | deficient, below           | hypotension - low blood pressure, hypodermic - below the skin                                                                                                              |
| in-       | inward, not                | inhalation - to breathe in, infertility - not fertile                                                                                                                      |
| Infra-    | beneath                    | infra-axillary - below the axilla                                                                                                                                          |
| inter-    | between                    | intervertebral - between the vertebrae                                                                                                                                     |
| intra-    | within                     | intramuscular - into the muscle                                                                                                                                            |
| juxta-    | near                       | juxta-articular - near a joint                                                                                                                                             |
| macro-    | large                      | macroglossia - an abnormally large tongue                                                                                                                                  |
| mal-      | bad, abnormal              | malformation - abnormally formed                                                                                                                                           |
| mega-     | great, large               | megacolon - enlarged colon                                                                                                                                                 |
| meta-     | change, beyond             | metaplasia - a change of one tissue to another, metastasis - the spreading of a malignant disease to distant parts of the body through the bloodstream or the lymph system |
| micro-    | small                      | microtia - having small ears                                                                                                                                               |
| mono-     | one                        | monochromatic - having only one colour                                                                                                                                     |
| morph- o- | shape                      | morphology - study of the form and structure of organisms                                                                                                                  |
| multi-    | many                       | Multigravida - a pregnant woman who has had more than one pregnancy                                                                                                        |
| neo-      | new                        | neonatal - pertaining to the first month of life.                                                                                                                          |
| nulli-    | none                       | nullipara - a woman who has never borne a child                                                                                                                            |
| olig- o-  | scanty, little             | oliguria - an abnormally low excretion of urine                                                                                                                            |
| pan-      | all                        | panacea - a medicine which is supposed to cure everything                                                                                                                  |
| Par- a-   | beside, abnormal           | paranasal - beside the nose, paraesthesia - abnormal or an unexplained tingling sensation                                                                                  |
| per-      | through                    | percutaneous - through the skin                                                                                                                                            |

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| peri-    | surrounding      | periosteum - membrane surrounding a bone                                                                                            |
| pico-    | one-trillionth   | picornavirus - extremely small RNA virus                                                                                            |
| poly-    | many, much       | polyuria - production of excessive amounts of urine                                                                                 |
| post-    | after            | postpartum - after childbirth                                                                                                       |
| pre-     | before           | precancer - a growth or cell which is not malignant but which may become cancerous                                                  |
| pseudo-  | false            | pseudocyesis - spurious or false pregnancy                                                                                          |
| quadri-  | four             | quadriplegia - paralysis of the four limbs of the body                                                                              |
| re-      | again            | reinfection - infection of an area for another time after recovery, especially with the same microorganism                          |
| retr- o- | backward, behind | retrograde - going backwards or deteriorating,<br>retroperitoneal - at the back of the peritoneum                                   |
| semi-    | Partial, half    | semicomatose - almost unconscious or half asleep, but capable of being woken up                                                     |
| sub-     | under, less than | subcutaneous - under the skin                                                                                                       |
| super-   | above            | supercilia - the eyebrow                                                                                                            |
| supra-   | above, upon      | suprarenal - above the kidneys                                                                                                      |
| syn-     | with, together   | syndrome - a group of symptoms occurring together regularly, and thus constituting a disease to which some particular name is given |
| tetra-   | four             | tetraplegia - paralysis of the body's four limbs, also called quadriplegia.                                                         |
| trans-   | through, across  | transdermal - entering through the skin, transurethral - across the urethra                                                         |
| un-      | not              | unconscious - not conscious                                                                                                         |
| uni-     | one              | unilateral - affecting one side of the body only                                                                                    |

| Medical Word Roots | Meanings        | Examples - Definitions                                                                                                       |
|--------------------|-----------------|------------------------------------------------------------------------------------------------------------------------------|
| abdomin- o-        | abdomen         | abdominoscopy - an internal examination of the abdomen, usually with an endoscope                                            |
| acou-              | hearing         | acoustics - the science of sounds                                                                                            |
| acr- o-            | extremity, peak | acrocyanosis - a blue coloration of the extremities, i.e. the fingers, toes, ears and nose, which is due to poor circulation |
| aden- o-           | gland           | adenocarcinoma - a malignant tumour of a gland                                                                               |
| adipo-             | fat             | adiposis - a state where too much fat is accumulated in the body                                                             |
| andr- o-           | male            | androsterone - one of the male sex hormones                                                                                  |
| angi- o-           | blood vessel    | angiosarcoma - a malignant tumour in a blood vessel                                                                          |
| ankyl              | crooked, fusion | ankylosis - a condition in which the bones of a joint fuse together                                                          |
| arteri- o-         | arteries        | arteriopathy - a disease of an artery                                                                                        |
| arthr- o-          | joint           | arthroplasty - a surgical operation to repair or replace a joint                                                             |
| bili               | bile            | biluria - the presence of bile in the urine                                                                                  |
| bi- o-             | life            | biocide - a substance which kills living organisms                                                                           |
| blast- or -blast   | embryonic state | blastocyst - an early stage in the development of an embryo                                                                  |
| blephar- o-        | eyelid          | blepharitis - inflammation of the eyelid                                                                                     |
| brachi- o-         | arm             | brachialis muscle - a muscle that causes the elbow to bend                                                                   |
| brady-             | slow            | bradypnoea - unusually slow breathing                                                                                        |
| calc- i            | heel            | calcaneus - the heel bone situated underneath the talus                                                                      |
| carcin- o-         | cancer          | carcinogen - a substance which produces a carcinoma or cancer                                                                |
| cardi- o-          | heart           | cardiopathy - any kind of heart disease                                                                                      |
| caud-              | tail            | caudal - toward the tail                                                                                                     |
| cephal- o-         | head            | cephalalgia - pain in the head                                                                                               |
| cerebr- o-         | cerebrum        | cerebrospinal - referring to the brain and the spinal cord                                                                   |

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| chem- o-                 | chemical   | chemotherapy - treatment with chemicals                                                                              |
| chol- e-                 | bile       | cholecystitis - acute or chronic inflammation of the gallbladder, causing severe abdominal pain                      |
| chondr- o-               | cartilage  | chondritis - inflammation of a cartilage                                                                             |
| col- i- o-               | colon      | colitis - inflammation of the colon                                                                                  |
| cost- o-                 | rib        | costalgia - pain around the chest due to damage to a rib or to one of the intercostal nerves beneath the ribs        |
| cut-                     | skin       | cutaneous - relating to the skin                                                                                     |
| cyan- o-                 | blue       | cyanosis - a bluish coloration of the skin or mucous membranes due to too much deoxygenated haemoglobin in the blood |
| cyst- i- o-              | bladder    | cystitis - inflammation of the urinary bladder, which makes someone pass water often and with a burning sensation    |
| cyt- o-                  | cell       | cytodiagnosis - diagnosis after examination of cells                                                                 |
| derm- or -derm           | skin       | dermatitis - inflammation of the skin                                                                                |
| dors- i- o-              | back       | dorsum - the back of any part of the body                                                                            |
| encephal- o-             | brain      | encephaloma - a tumour of the brain                                                                                  |
| enter- o-                | intestine  | enterocolitis - inflammation of the colon and small intestine                                                        |
| erythr- o-               | red        | erythrocytes - a mature red blood cell                                                                               |
| fasci-                   | bundle     | fasciae - bundles of muscle fibers                                                                                   |
| feбри-                   | fever      | febrile - feverish or related to fever, as in febrile convulsions                                                    |
| fil-                     | threadlike | filariasis - a tropical disease caused by parasitic threadworms in the lymph system, transmitted by mosquito bites   |
| galact- o-               | milk       | galactorrhoea - the excessive production of milk                                                                     |
| gastro-                  | stomach    | gastritis - inflammation of the stomach                                                                              |
| ger- o- or geront- o-    | aging      | gerontology - study of aging                                                                                         |
| gloss- o- a-             | tongue     | glossitis - inflammation of the surface of the tongue                                                                |
| haeme- a- o- or haemato- | blood      | haematology - the scientific study of blood, its formation and its diseases                                          |
| hepat- o-                | liver      | hepatitis - inflammation of the liver through disease or drugs                                                       |

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|-------------|--------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------|
| hist- i- o- | tissue                   | histogenesis - the formation and development of tissue from the embryological germ layer                                                             |
| hydro-      | water                    | hydrorrhoea - a discharge of watery fluid                                                                                                            |
| hyster- o-  | uterus                   | hysterotomy - a surgical incision into the uterus, as in caesarean section or for some types of abortion                                             |
| ile- o-     | ileum                    | ileostomy - a surgical operation to make an opening between the ileum and the abdominal wall to act as an artificial opening for excretion of faeces |
| ischi- o-   | hip                      | ischiopubic - pertaining to the ischium and pubes                                                                                                    |
| jejun- o-   | jejunum                  | jejunectomy - a surgical operation to cut into the jejunum                                                                                           |
| kerat- o-   | horny tissue, cornea     | keratoma - a hard-thickened growth due to hypertrophy of the horny zone of the skin                                                                  |
| kine- t- o- | movement                 | kinesiology - the study of human movements, particularly with regard to their use in treatment                                                       |
| labio-      | lips, labia              | labioplasty - a surgical operation to repair damaged or deformed lips                                                                                |
| lact- o-    | milk                     | lactose - a type of sugar found in milk                                                                                                              |
| laryng- o-  | larynx                   | laryngology - the study of diseases of the larynx, throat and vocal cords                                                                            |
| leuk- o-    | white                    | leukocytes - a white blood cell which contains a nucleus but has no haemoglobin                                                                      |
| lip- o-     | fat                      | lipoma - a benign tumour formed of fatty tissue                                                                                                      |
| lith- o-    | stone                    | litholapaxy - the evacuation of pieces of a stone in the bladder after crushing it with a lithotrite                                                 |
| mamm- o-    | breast                   | mammoplasty - plastic surgery to alter the shape or size of the breasts                                                                              |
| mast- o-    | breast                   | mastectomy - the surgical removal of a breast                                                                                                        |
| melan- o-   | black                    | melanoma - a tumour formed of dark pigmented cells                                                                                                   |
| meno-       | menses                   | menorrhagia - very heavy bleeding during menstruation                                                                                                |
| my- o-      | muscle                   | myocele - a condition in which a muscle pushes through a gap in the surrounding membrane                                                             |
| myc- o-     | fungus                   | mycosis - any disease caused by a fungus, e.g. athlete's foot                                                                                        |
| myel- o-    | bone marrow, spinal cord | myelocyte - a cell in bone marrow which develops into a granulocyte                                                                                  |
| myx         | mucus                    | myxoma - a benign tumour of mucous tissue, usually found in subcutaneous tissue of the limbs and neck                                                |
| nas- o-     | nose                     | nasosinusitis - a condition in which the nose and sinuses swell up                                                                                   |

|                  |                         |                                                                                                          |
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| neph- o-         | kidney                  | nephrolithiasis - a condition in which stones form in the kidney                                         |
| neur- o-         | nerve or nervous system | neurohormone - a hormone produced in some nerve cells and secreted from the nerve endings                |
| ocul- o-         | eye                     | oculoplethysmography - measurement of the pressure inside the eyeball                                    |
| onc- o-          | tumor - cancerous       | oncogene - a part of the genetic system which causes malignant tumours to develop                        |
| ophthalm- o-     | eye                     | ophthalmoplegia - paralysis of the muscles of the eye                                                    |
| orchi- o-        | testes                  | orchitis - inflammation of the testes, characterised by hypertrophy, pain and a sensation of weight      |
| oro              | mouth                   | oropharynx - a part of the pharynx below the soft palate at the back of the mouth                        |
| oste- o-         | bone                    | osteitis - inflammation of a bone due to injury or infection                                             |
| ot- o-           | ear                     | otorrhoea - the discharge of pus from the ear                                                            |
| paed- or ped- o- | child                   | paediatrics - the study of children, their development and diseases.                                     |
| path- o-         | disease                 | pathogen - microorganism which causes a disease                                                          |
| Pharmaco-        | drugs                   | pharmacokinetics - the study of how the body reacts to drugs over a period of time                       |
| pharyng- o-      | pharynx                 | pharyngitis - sore throat                                                                                |
| phleb- o-        | vein                    | phlebolith - a stone which forms in a vein as a result of an old thrombus becoming calcified             |
| phot- o-         | light                   | photopsia - a condition of the eye in which someone sees flashes of light                                |
| plasm- o-        | liquid part of blood    | plasminogen - a substance in blood plasma which becomes activated and forms plasmin                      |
| pleur- o-        | pleura, rib, side       | pleurisy - inflammation of the pleura                                                                    |
| pneum- o-        | lung                    | pneumonia - inflammation of a lung, where the tiny alveoli of the lung become filled with fluid          |
| pod- o-          | foot                    | podiatry - the study of minor diseases and disorders of the feet                                         |
| proct- o-        | anus or rectum          | proctoclysis - the introduction of a lot of fluid into the rectum slowly                                 |
| psych- o-        | mind                    | psychiatry - study and treatment of mental disorders                                                     |
| pyel- o-         | kidney                  | pyelotomy - a surgical operation to make an opening in the pelvis of the kidney                          |
| pyr- o-          | burning or fever        | pyrogen - a substance which causes fever                                                                 |
| ren- o-          | kidney                  | renography - an examination of a kidney after injection of a radioactive substance, using a gamma camera |

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|--------------------|----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| rhin- o-           | nose                 | rhinoplasty - plastic surgery to correct the appearance of the nose                                                                                                                                                          |
| sarc- o-           | flesh                | sarcoma - a highly malignant tumour made of connective tissue cells                                                                                                                                                          |
| scler- o-          | hard                 | scleroma - a patch of hard skin or hard mucous membrane                                                                                                                                                                      |
| soma- or somat- o- | body                 | somatization - psychiatric condition expressed through physical symptoms                                                                                                                                                     |
| sten- o-           | narrow               | stenosis - a condition in which a passage becomes narrow                                                                                                                                                                     |
| tachy              | Rapid, fast          | tachyarrhythmia - a fast irregular heartbeat                                                                                                                                                                                 |
| therm- o-          | heat                 | thermoanaesthesia - a condition in which someone cannot tell the difference between hot and cold                                                                                                                             |
| thorac- o-         | chest                | thoracotomy - a surgical operation to remove one or more ribs                                                                                                                                                                |
| thromb- o-         | clot                 | thromboangiitis - a condition in which the blood vessels swell and develop blood clots along their walls                                                                                                                     |
| trache- o-         | trachea              | tracheobronchitis - inflammation of both the trachea and the bronchi                                                                                                                                                         |
| ur- o-             | urinary, urine       | urochesia - the passing of urine through the rectum, due to injury of the urinary system                                                                                                                                     |
| vas- o-            | Vessel, vas deferens | vasopressor - a substance which increases blood pressure by narrowing the blood vessels, vasoligation - a surgical operation to tie the vasa deferentia to prevent infection entering the epididymis from the urinary system |
| ven- i- o-         | vein                 | venoclysis - the procedure of slowly introducing a saline or other solution into a vein                                                                                                                                      |
| vesic- o-          | bladder              | vesicospinal - pertaining to the urinary bladder and spine                                                                                                                                                                   |



| Suffix     | Meaning          | Example - Definition                                                                                     |
|------------|------------------|----------------------------------------------------------------------------------------------------------|
| -algia     | pain             | neuralgia - a spasm of pain which runs along a nerve                                                     |
| -cyte      | cell             | leucocyte - white blood cell                                                                             |
| -ectasis   | dilation         | bronchiectasis - dilated bronchi                                                                         |
| -ectomy    | surgical removal | splenectomy - a surgical operation to remove the spleen                                                  |
| -oedema    | excessive fluid  | angioedema - fluid buildup that causes swelling under the skin                                           |
| -emesis    | vomiting         | hematemesis - vomiting of blood                                                                          |
| -itis      | inflammation     | colitis - inflammation of the colon                                                                      |
| -ism       | state of         | hypothyroidism - state of low thyroid                                                                    |
| -logist    | one who studies  | cardiologist - a doctor who specialises in the study of the heart                                        |
| -lys- i- o | breakdown        | cytolysis - the breaking down of cells                                                                   |
| -oma       | tumour           | glioma - any tumour of the glial tissue in the brain or spinal cord                                      |
| -osis      | condition        | fibrosis - the process of replacing damaged tissue by scar tissue                                        |
| -pathy     | disease          | myopathy - muscle disease                                                                                |
| -plasia    | growth           | hypoplasia - a lack of development or incorrect formation of a body tissue or an organ                   |
| -plasty    | surgical repair  | angioplasty - plastic surgery to repair a blood vessel, e.g. a narrowed coronary artery                  |
| -plegia    | paralysis        | hemiplegia - severe paralysis affecting one side of the body due to damage of the central nervous system |
| -pnoea     | breathing        | orthopnoea - a condition in which a person has great difficulty in breathing while lying down            |
| -poiesis   | production       | erythropoiesis - the formation of red blood cells in red bone marrow                                     |
| -praxia    | movement         | apraxia - a condition in which someone is unable to make proper movements                                |
| -rrhoea    | fluid discharge  | diarrhoea - a condition in which someone frequently passes liquid faeces                                 |
| -scope     | observe          | endoscope - tool for observing the interior of body organs                                               |
| -stomy     | opening          | duodenostomy - a permanent opening made between the duodenum and the abdominal wall                      |
| -taxis     | movement         | ataxia - a failure of the brain to control movements                                                     |

|         |          |                                                                                                                                      |
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| -tomy   | incision | canaliculotomy - a surgical operation to open up a little canal                                                                      |
| -tripsy | crushing | lithotripsy - the process of breaking up kidney or gall bladder stones into small fragments that the body can eliminate them unaided |
| -trophy | growth   | hypertrophy - an increase in the number or size of cells in a tissue                                                                 |